

rules, are necessary. In an electronic environment programmatically embedded instructions can verify compliance with the coding rules.

Electronic

1. Enter the statement or term in the coding tool
2. Select the matching term, or the one closest to what you are looking for from amongst the displayed options
3. Verify the result in the tabular list browser view for exclusions, inclusions and notes given at the level of that category, its grouping levels and at the chapter level.

Paper

1. Look up the lead term in the Alphabetical Index and applicable secondary terms.
2. Select the appropriate term, or one closest to what you are looking for, from amongst the listed options
3. Verify the result in the tabular list (Volume 1) for exclusions, inclusions and notes given at the level of that category, its grouping levels and at the chapter level.

The WHO online browser and coding tool are available at <https://icd.who.int>.

2.7 ICD–11 conventions

ICD–11 has standard ways of presenting its content. Conventions describe textual content and also apply to the coding structure.

2.7.1 Inclusions

Within the coded categories there are typically other optional diagnostic terms. These are known as ‘inclusion terms’ and are given, in addition to the title, as examples of the diagnostic statements to be classified to that category. They may refer to different conditions or be synonyms. They are not a sub-classification of the category.

Inclusion terms are listed primarily as a guide to the content of the category, in addition to the descriptions. Many of the items listed relate to important or common terms belonging to the category. Others are borderline conditions or sites listed to distinguish the boundary between one subcategory and another. The lists of inclusion terms are by no means exhaustive.

Alternative names of diagnostic entities (synonyms) are included and shown in the electronic coding tool and the Alphabetical Index.

It is sometimes necessary to read inclusion terms in conjunction with titles. This usually occurs when the inclusion terms describe lists of sites or pharmaceutical products, where appropriate words from the title (e.g. ‘malignant neoplasm of ...’, ‘injury to ...’, ‘toxic effects of ...’) need to be understood. General diagnostic descriptions common to a range of categories, or to all the subcategories in a four-character category, are to be found in the notes heading ‘Inclusions’, immediately following a chapter, group, or category title.

2.7.2 Exclusions

Certain categories contain lists of conditions preceded by the word ‘Exclusions’. These are terms which are classified elsewhere. An example of this is [5A60](#) *Hyperfunction of pituitary gland* which excludes Cushing syndrome.

Exclusions serve as a cross reference in ICD and help to delineate the boundaries of a category.

General exclusions for a range of categories or for all subcategories are found in the notes heading 'Excludes', immediately following a chapter, group or category title.

Multiple parenting in ICD-11 shows categories in the context of siblings that are placed elsewhere in the classification. This is also an indication of an exclusion and means 'a sibling is coded elsewhere'. In the print and the electronic version this is marked with the label 'code elsewhere'.

2.7.2.1 'Code also' and 'Use additional code, if desired' instructions

'Code also' instructions inform the user about required additional aetiological information which is mandatory to be coded in a cluster with certain categories because that additional information is relevant for primary tabulation. The 'code also' statement marks the categories that must be used in conjunction with the indicated second code(s). However, in some instances aetiology may be unknown although the condition requires treatment in its own right. In this circumstance, the code may be reported alone.

For example, the category Diabetic cataract indicates 'code also' type of diabetes. This means that in conjunction with the code for 'diabetic cataract', the code for the type of diabetes should be assigned. Both stem codes for the type of diabetes and the diabetic cataract are always reported in a cluster.

'Use additional code, if desired' - instructions inform the user about optional additional detail that can be coded.

2.7.3 'NEC' and 'NOS'

2.7.3.1 'NEC'

The stem 'not elsewhere classified', when used in a category title, serve as a warning that certain specified variants of the listed conditions may appear in other parts of the classification. For example, [NF09](#) Adverse effects, not elsewhere classified. Codes to which the NEC description is appended should only be used if one of the other options available in the classification is not suitable.

2.7.3.2 'NOS'

The letters NOS are an abbreviation for the term 'not otherwise specified', implying that the documentation that is used for classifying does not provide more detail beyond the term provided. It implies 'unspecified', 'incompletely specified' or 'unqualified'. Sometimes an unqualified term is nevertheless classified to a rubric for a more specific type of the condition. This is because, in medical terminology, the most common form of a condition is often known by the generic name of the condition itself and only the less common types are qualified. For example, 'pharyngitis' is commonly used to mean 'acute pharyngitis'. These inbuilt assumptions have been taken into account in order to avoid incorrect classification.

Careful inspection of inclusion terms will reveal where an assumption of cause has been accounted for. Coders should be careful not to code a term as unqualified unless it is quite clear that no information is available that would permit a more specific assignment

elsewhere. Similarly, in interpreting statistics based on the ICD, some conditions assigned to an apparently specified category will not have been so specified on the record that was coded. When comparing trends over time and interpreting statistics, it is important to be aware that assumptions may change from one revision of the ICD to another. For example, before the Eighth Revision, an unqualified aortic aneurysm was assumed to be due to syphilis (this is no longer the case since the introduction of ICD–10). In ICD-11 in most instances the ‘NOS’ terminology points to unspecified categories, so that future data analysis can take care of assumptions regarding the linguistic meaning.

Additional terms permitted in ICD coding:

- Certain
- Other
- Unspecified
- And
- Or
- Due to
- With
- Caused by
- Attributed to
- Secondary to
- Associated with

2.7.4 ‘Certain’

The term ‘certain’ is used where certain entities that could be grouped in a specific location in the classification are grouped somewhere else outside the current chapter or block. For example, [8B22](#) *Certain specified cerebrovascular diseases* means that only some specified cerebrovascular diseases are coded here, whereas other specific types of cerebrovascular disease are located elsewhere in the classification.

2.7.5 Residual categories – ‘Other’ and ‘Unspecified’

ICD-11 coding should always be completed to include the most specific level of detail possible with the use of one code or multiple codes as described above. There are, however, circumstances when that is not possible and for that reason the ICD-11 includes categories titled ‘other’ and ‘unspecified’. In some instances, necessary information to select a specific category may not be available in the source documentation. When this is the case, the residual category ‘unspecified’ is selected. Conversely, there are instances where the information in the source documentation is very specific, but the tabular list does not include a specific category. In this case, users identify the closest category match, and code to the residual category titled ‘other’.

2.7.6 Use of ‘And’ and ‘Or’

The words ‘and’ and ‘or’ in ICD–11 are used in their meaning in formal logic. A term that includes a statement of the kind ‘A and B’ means that both A and B, have to be present in order to use that category. A term that includes a statement of the kind ‘A or B’ means that the category may be used if either A or B are present.

2.7.7 'Due to' and 'Associated with'

'Due to' is the preferred term for categories where two conditions are mentioned, and a causal sequence exists between them. Other terms, such as 'caused by' or 'attributed to' are allowable synonyms. The phrase 'secondary to' is equivalent and may also be included as a synonym. 'Associated with' is the preferred term for categories where two conditions are mentioned but there is no causal sequence implied.

2.7.8 Spelling, parentheses, grammar and other conventions

Spelling and grammar of ICD-11 follow the British rules with exceptions and amendments conforming to WHO spelling rules. The detailed conventions are listed below. The ICD-11 terminology uses the following conventions:

- Terms are listed in their singular form. For example, 'Superficial injury of scalp' instead of 'Superficial injuries of scalp'. There are exceptions when the singular form doesn't apply, such as 'Multiple injuries of head', and when a term describes a group of diseases, such as "Benign vascular neoplasms".
- No use of apostrophes with eponyms. For example, 'Hodgkin lymphoma' (instead of 'Hodgkin's lymphoma')
- Entities are described using natural language. For example, 'myocardial infarction' (instead of 'infarction, myocardial').
- Abbreviations are printed using upper case letters and followed by the complete title in full. For example, 'MI – [myocardial infarction]'.
- Parentheses are used in the tabular list to enclose the code to which an exclusion term refers. For example, [9A01.3](#) *Infectious blepharitis* Exclusions: Blepharoconjunctivitis ([9A60.4](#))

2.7.9 General features

The main structural innovation of ICD-11 is that it is built on a Foundation component from which the tabular lists (such as the classification for morbidity and mortality statistics) is derived. See [1.2.5 Foundation Component and Tabular lists of ICD-11](#).

Table 1: *ICD-11 Terminology*

ICD-11 Term	Explanation
Foundation component	Underlying data base content that holds all necessary information to generate print versions of the tabular list and the alphabetical index, as well as additional information that is needed to generate specialty linearisations of ICD-11 and country specific modifications.
Stem code	Stem codes are codes that can be used alone. They are found in the tabular list of ICD-11 for Mortality and Morbidity Statistics. Stem codes may be entities or groupings of high relevance, or clinical conditions that should always be described as one single category. The design of stem codes makes sure that in use cases that require only one code per case, a meaningful minimum of information is collected.
Extension code	Extension codes are lists of additional information that can be added to a stem code when users and settings are interested in reporting more detail. Extension codes are not mutually exclusive. They are not designed to be a classification but may show hierarchies and can never be used without a stem code in the context of statistical classification. Extension codes can never appear in the first position in a cluster. Extension codes may be used alone in another context, e.g. for device documentation.
Precoordination	Stem codes may contain all pertinent information about a clinical concept in a pre-combined fashion. This is referred to as 'precoordination'. Example: BD50.40 <i>Abdominal aortic aneurysm with perforation</i> Example: CA40.04 <i>Pneumonia due to Mycoplasma pneumoniae</i>
Postcoordination	The use of multiple codes (i.e. stem codes and/or extension codes) together to fully describe a documented clinical concept. Postcoordination is by matter of principle permitted with all axes, as long as at least one stem code is used. API, Browser and coding tool facilitate postcoordination in certain areas. Other areas of postcoordination have to be coded individually, but may be supported by future updates of ICD-11, where need arises.
Cluster	A cluster is the postcoordinated entities that are joined using either a forward slash (/) or ampersand (&). Example: Diagnosis: Duodenal ulcer with acute haemorrhage Cluster: DA63.Z / ME24.90 Condition - DA63 <i>Duodenal ulcer, unspecified</i> Has manifestation (use additional code, if desired) - ME24.90 <i>Acute gastrointestinal bleeding, not elsewhere classified</i>

ICD-11 Term	Explanation
Primary and secondary parents	The hierarchy of ICD-11 is defined the same as it was in previous versions of ICD. The ability to connect specific diseases and concepts within the classification to another parent code was introduced to enable specific extracts of the Tabular list for medical specialties or for specific use cases. Example: A code for a malignant neoplasm of the skin is in the chapter for malignant neoplasms. The primary parent for this code is a code or a block from this chapter. However, a medical doctor treating only skin diseases might want to see only codes from the classification that are relevant for his or her specific clinical purpose. Therefore, a secondary parent was defined in the skin chapter which will only show the code in this chapter if the specific extract of codes for his or her use case is selected.

2.8 Stem codes

ICD-11 stem codes are codes in a particular tabular list that can be used alone. Stem codes may be entities or groupings of high relevance, or clinical entities that should always be described as one entity. The design of stem codes makes sure that in use cases that require only one code per case a meaningful minimum of information is collected.

The stem codes of the ICD-11 are organised in 26 chapters that follow the traditional pattern of the ICD, relating to aetiology, relevant organ system, maternal status, perinatal status, external causes, and factors influencing health status.

2.9 Extension codes

Extension codes are provided for use as supplementary or additional codes when it is desired to identify more detail in classification entities elsewhere. The inclusion of the new Extension codes in ICD-11 provides capacity for coding qualifying information and are linked to stem codes. Extension codes have been designed to standardise the way additional information is added to stem codes, and the adoption of multi-dimensional coding results in a substantially reduced amount of stem codes. Extension codes may be used alone in other contexts, e.g. device documentation.

Extension codes are not mutually exclusive. They are not a classification and can never be used without a stem code for statistical purposes. Extension codes can never appear in the first position in a classification cluster. One or more extension codes can be linked when coding a specific condition.

There are two main types of Extension codes:

- Type 1 extension codes allow the user to add detail to a stem code in terms of severity, temporality, anatomy, histopathology of the condition or other dimensions like substances and medical devices. For example, if the diagnostic statement reads 'cervical disc prolapse C5-C6' the anatomy extension code [XA1X49](#) *Cervical intervertebral disc or space C5-C6* can be added to the stem code [FA80.1](#) *Intervertebral disc degeneration of cervical spine with prolapsed disc* in order to capture the detail of contained in the diagnostic statement.

- Type 2 extension codes represent diagnosis code descriptors which indicate how the diagnosis is to be used and/or interpreted. The meaning of the code refers to the same condition, but the use of type 2 – diagnosis code descriptor extension code alters its interpretation. For example, for adverse event reporting it is important to code diagnosis timing in terms of [XY6M](#) *Present on admission*, [XY69](#) *Developed after admission*, or [XY85](#) *Uncertain timing of onset relative to admission*.

Overview of the Type 1 Extension codes

- Severity scale value
- Temporality (course of the condition)
- Aetiology
- Topology Scale Value
- Anatomy and topography
- Histopathology
- Dimensions of injury
- Dimensions of external causes
- Consciousness
- Substances
- ICD-O
- Health Devices, Equipment and Supplies

Overview of Type 2 – Extension codes - Diagnosis Code Descriptors

- Discharge diagnosis types
- Diagnosis timing
- Diagnosis timing in relation to surgical procedure
- Diagnosis method of confirmation
- Diagnosis certainty
- Obstetrical diagnosis timing
- Encounter descriptors
- Capacity or context

2.10 Precoordination and postcoordination

Some stem codes contain all pertinent information about a clinical concept in a pre-combined fashion. This is referred to as ‘precoordination’.

A health condition may be further described to any level of detail, by applying more than one code, or by ‘postcoordinating’ (i.e. combining codes):

- two or more stem codes, (i.e. code1/code2)
- stem codes with one or more extension codes. (i.e. stem code&extension code1&extension code2)

A group of codes that have been postcoordinated is called a ‘cluster’. A forward slash (/) or ampersand (&) is used to show the linkage between postcoordinated codes. In this manner, the classification can address many clinical concepts with a limited range of categories.

Example

Precoordination of concepts in a single code Condition: [2C25.2 Squamous cell carcinoma of bronchus or lung](#) has precoordination, both site and pathology are combined in a single precoordinated stem code.

Example

Postcoordination of concepts combined in a cluster: the condition urinary tract infection due to Extended spectrum beta-lactamase producing *Escherichia coli* is expressed through a combination of two linked or postcoordinated stem codes. Condition: [GC08.0 Urinary tract infection, site not specified, due to Escherichia coli](#) Associated with (use additional code, if desired): [MG50.27 Extended spectrum beta-lactamase producing Escherichia coli](#) Cluster code: [GC08.0/MG50.27](#)

Postcoordination axis

‘Has causing condition’ - this field is indicating the causing condition that must be coded when known. The causing condition can be compared to the ‘dagger’ code in ICD-10. This option is found at entities that are typically caused by a range of different conditions and is referred to as mandatory postcoordination in the Coding tool.

‘Has manifestation’ - prompts the user to code any manifestations. Manifestations can be compared to the ‘asterisk’ codes in ICD-10. This option is found at entities that can develop manifestations.

‘Associated with’ - when conditions are captured together for a full picture but do not necessarily represent a cause and effect scenario.

Description

It is mandatory to code the causing condition for primary tabulation when it is known. ‘Has causing condition’ is added to categories that are caused by an underlying disease. For example, retinopathy has a ‘causing condition’ of diabetes. Causing conditions should be considered required in almost all situations, and must be used for conditions that are manifestations.

It is optional to code manifestations of a disease. For example, diabetes ‘Has manifestations’ such as retinopathy. This coding should be considered ‘Allowed’ in almost all situations. The listed manifestations are usually a sample of the ones frequently resulting from the condition.

This field is used when multiple codes are required to fully describe a condition. For example, ‘Associated with’ is used to link the codes for antimicrobial resistance to the codes for the infection. This coding can be either ‘Allowed’ or ‘Required’ depending on the situation.

Special cases for postcoordination:

1. External cause codes are 'allowed' to identify the cause of an injury and are 'associated with' the injuries.
2. The external cause code to identify a specific drug is 'allowed' with entities beginning with or including the phrase 'Drug -induced'.
3. External cause codes for the mode and mechanism of health care related harm are 'associated with' the codes for the harm.

2.10.1 Adding detail – postcoordination and cluster coding with multiple stem codes and extension codes

Information about the aetiology and the manifestation of the condition of interest should be coded. In some instances, the ICD code refers to both (i.e. precoordinated), while in other instances more than one stem code (and/or extension code) needs to be used in order to express the relevant detail. This requires postcoordination.

Note: When a condition is already precoordinated and given an ICD-11 MMS code, the existing precoordinated ICD-11 MMS stem code must be used. Postcoordination must never be used to replicate the meaning of a precoordinated code.

Also note that all ICD entities in the Foundation have a uniform resource identifier (URI). If a particular entity does not have an ICD-11 MMS code, and collection of a more detailed/specific entity is required for selected national needs, a country may collect the URI for this purpose. Refer to section 2.12 Electronic recording and reporting.

E.g. Acute bleeding duodenal ulcer

Stem Code: [DA63.Z](#) Duodenal ulcer, unspecified Has manifestation (use additional code, if desired): [ME24.90](#) Acute gastrointestinal bleeding, not elsewhere classified Cluster: [DA63.Z/ME24.90](#)

E.g. Acute RSV bronchiolitis

Code: [CA41.0](#) Acute bronchiolitis due to respiratory syncytial virus Explanation: Since RSV bronchiolitis is a precoordinated concept in ICD-11, it is incorrect/prohibited to replicate the meaning of the diagnostic statement using a stem code and extension code (i.e. do not code: [CA41.Z] Acute bronchiolitis, unspecified&[XN275](#) *Human respiratory syncytial virus*)

E.g. Fracture, shaft of ulna

Code: [NC32.2](#) Fracture of shaft of ulna Explanation: Since fracture of shaft of ulna is a precoordinated concept in ICD-11, it is incorrect/prohibited to replicate the meaning of the diagnostic statement using a stem code and extension code (i.e. do not code: [NC32.Z](#) Fracture of forearm, unspecified&[XA8U33](#) shaft of the ulna)

There may be less obvious cases across the ICD. In an electronic environment, programmatically embedded instructions will help to avoid this kind of mistake. For reporting purposes, any correlated codes are linked using a forward slash (/) between stem codes and an ampersand (&) to separate stem codes with extension codes.

2.10.2 Combining stem codes and extension codes, and how to order these in a complex code cluster

Stem codes from other parts of ICD and extension codes can be linked together to describe a clinical concept in detail. They have to be grouped together in data transmission and evaluation in order to not lose the information conveyed by the joint group of codes. Such a

group of codes is called a cluster. Cluster coding requires use of a specific syntax to indicate which codes belong together when postcoordination is used. This syntax has to comply with the following rules:

1. If only one stem code is coded, no clustering mechanisms need to be observed.

E.g. Condition: Acute ST elevation myocardial infarction [BA41.0](#) *Acute ST elevation myocardial infarction*
2. When postcoordinating to form a cluster, stem codes are always coded before extension codes. (Note, however, the complex clustering scenario depicted in Example 5 below, where a combination of multiple stem codes and linked extension codes are combined in a single complex cluster).
3. If one stem code is postcoordinated with one or more extension codes, the combining syntax used is the ampersand (&).

Example 1: Acute ST elevation myocardial infarction, anterior wall, LAD

Condition (code) - Acute ST elevation myocardial infarction [BA41.0](#) *Acute ST elevation myocardial infarction* Specific anatomy - [XA7RE3](#) *Anterior wall of heart*
Specific anatomy - [XA7NQ7](#) *Left anterior descending coronary artery* **Cluster:** [BA41.0&XA7RE3&XA7NQ7](#)

Example 2: Acute pyelonephritis, left side, E. coli Condition (code) - [GB51](#) *Acute pyelonephritis*

Laterality - [XK8G](#) *Left* Infectious agent - [XN6P4](#) *Escherichia coli* **Cluster:** [GB51&XK8G&XN6P4](#)

4. If two stem codes are postcoordinated to provide additional detail, it is important to follow the order (within a cluster) according to the use case (e.g. mortality or morbidity). The first stem code will be separated from the second stem code by a slash (/).

If only one code can be retained during data analysis for mortality (underlying cause of death) and public health prevention, priority of order should be given to the code that best describes the aetiology of a condition. If only one code can be retained for morbidity data analysis, priority should be given to the main condition (reason for admission after study established at the end of the episode of health care).

Example 3: Mortality (underlying cause of death) code ordering within a cluster

Patient died because of their diabetic coma. The patient had Type 2 diabetes mellitus. Condition (terminal cause of death): [5A23](#) *Diabetic coma* Condition (underlying cause of death): [5A11](#) *Type 2 diabetes mellitus*
Mortality cluster order: [5A11/5A23](#)

Example 4: Morbidity (main condition) code ordering within a cluster (if only one code can be retained during data analysis)

Patient admitted to hospital in a diabetic coma. The patient had Type 2 diabetes mellitus.
Main condition: [5A23](#) *Diabetic coma* Other condition: [5A11](#) *Type 2 diabetes mellitus* **Morbidity cluster order:** [5A23/5A11](#)

5. If a stem code is postcoordinated with extension codes and another stem code with some more extension codes is also coded within a cluster, the specific syntax should be designed to make a clear distinction between which extension codes in the cluster belong to which stem codes. The following syntax must be followed: The first stem code is reported, followed by an ‘&’ followed by one or more extension codes,

each of them separated by '&'. Then a slash '/' separates this first section of the cluster from the next stem code which is followed by '&' and the extension codes for this specific stem code, each again separated by '&'.

Example 5: stem code & extension code / stem code & extension code & extension code

Left inguinal hernia with acute obstruction Condition (code) - [DD51](#) *Inguinal hernia*
Laterality - [XK8G](#) *Left*
Has manifestation (use additional code, if desired) - [ME24.2](#) *Digestive system obstruction*
Temporal pattern and onset - [XT5R](#) *Acute* **Cluster:** [DD51&XK8G/ME24.2&XT5R](#)

Postcoordination is only to be used to combine codes to describe and fully characterise a documented clinical concept. If the documentation describes two distinct clinical concepts that are represented by separate stem codes, they should not be reported together in a postcoordinated cluster.

Example 6: Pedestrian fall injury

Concussion and open fracture shaft of left ulna due to fall on uneven sidewalk:

Condition (code) 1 - [NA07.0Z](#) *Concussion, unspecified*
Associated with (use additional code, if desired) - [PA60](#) *Unintentional fall on the same level or from less than 1 metre*
Object or substance producing injury - [XE1DA](#) *Uneven surface, not elsewhere classified*
Place of occurrence - [XE53A](#) *Sidewalk* **Cluster:** [NA07.0/PA60&XE1DA&XE53A](#)
Condition (code) 2 - [NC32.2](#) *Fracture of shaft of ulna*
Laterality - [XK8G](#) *Left*
Fracture open or closed - [XJ7YM](#) *Open fracture*
Associated with: [PA60](#) *Unintentional fall on the same level or from less than 1 metre*
Objects of living things involved in causing - [XE1DA](#) *Uneven surface, not elsewhere classified*
Place of occurrence - [XE53A](#) *Sidewalk* **Cluster:** [NC32.2 & XK8G&XJ7YM/PA60 & XE1DA & XE53A](#)

Programmatically embedded instructions in the Foundation component of ICD-11 facilitate use of frequently used code combinations. Code combinations are not limited to the ones facilitated by these instructions. Additional instructions are added, based on user demands.

2.10.3 Diagnosis Timing - 'Present on admission' vs. 'Developed after admission'

Among the new Type 2 Extension codes -Diagnosis Code Descriptors, diagnosis timing is the particularly important set of extension codes that allow for distinction of diagnoses present on admission from diagnoses arising after admission, i.e. during the period of hospitalisation.

The latter distinction is particularly important, because it allows for the targeted identification of a number of in-hospital diagnoses that may represent adverse events associated with health care. The majority of coded concepts in a hospital record are conditions present on admission. Recognising this, it will be of significant interest to flag a diagnosis that developed after admission.

Example 1:

A patient with long-standing type 1 diabetes, admitted to hospital because of a myocardial infarction.
Main condition: Myocardial infarction
Other condition: Diabetes mellitus, type 1

In this instance, both conditions are present at admission, but one of them (myocardial infarction) does not need to be coded as being 'present on admission' because it is the main condition, designated in this example as being 'the condition that is determined to be the reason for admission, established at the end of the episode of health care'. The appropriate coding of this scenario therefore includes two clusters, each of which involves a stem code linked to an accompanying extension code i.e.:

- 'Stem code for acute myocardial infarction' & 'Discharge Diagnosis Type Extension code for main condition'; [BA41.Z](#) & [XY0Y](#)
- 'Stem code for diabetes mellitus type 1' & 'Diagnosis timing Extension code for present on admission'; [5A10](#) & [XY6M](#)

Note that for both coded entities in the above example, an ampersand (&) is used. In the first cluster, the stem code for myocardial infarction is linked to a diagnosis type extension code for main condition diagnosis type. In the second cluster, the stem code for diabetes mellitus type 1 is linked to a diagnosis timing extension code for present on admission.

Example 2:

A patient with long-standing type 1 diabetes, admitted to hospital because of chest pain. After assessment diagnosed with myocardial infarction. The patient develops deep vein thrombosis in right lower limb as an in-hospital complication of care.

Main condition: Myocardial infarction

Other conditions: Diabetes mellitus, type 1; Deep vein thrombosis (arising after hospital stay began)

In this example, a diagnosis timing extension code for 'developed after admission' is linked by cluster coding to a stem code for 'deep vein thrombosis'. The first two diagnostic concepts, meanwhile, are coded exactly as per the preceding example. i.e.

- 'Stem code for acute myocardial infarction' & 'Discharge Diagnosis Type Extension code for main condition'; [BA41](#) & [XY0Y](#)
- 'Stem code for diabetes mellitus type 1' & 'Diagnosis Timing Extension code for present on admission'; [5A10](#) & [XY6M](#)
- 'Stem code for lower limb deep vein thrombosis' & 'Right' & 'Diagnosis Timing Extension code for developed after admission'; [BD71.4](#) & [XK9K](#) & [XY69](#)

Again, each of the three cluster entities uses an ampersand '&' because the second code (and third code) in the cluster is an extension code.

2.11 Functioning section

The Functioning section of ICD-11 allows coding and assessment of functioning in line with ICF but at an operational level. For detailed recording and assessment, the full ICF should be used. This section's concepts are aligned with ICF and allow an easy transition to ICF. The design of the functioning section in ICD-11 addresses documentation and assessment of the level of functioning of persons, for

- general medical practice, as work incapacity assessment
- social benefits as for disability, or accident pension
- payment or reimbursement purposes
- needs assessment as in rehabilitation, occupational assistance, or long term care
- outcome evaluation of treatment

The functioning section in ICD-11 provides clinician-friendly tools for standardized assessment and documentation of functioning. The functioning section of ICD-11 allows for adding future additional assessment instruments.

2.11.1 Functioning assessment

2.11.1.1 WHO DAS 2.0: features and use cases

The WHO Disability Assessment Schedule (WHO-DAS 2.0) is the internationally, culturally and socioeconomically validated generic functioning assessment instrument for an adult population. WHO DAS 2.0 is focused on ICF Activity and Participation domains and allows to compute an overall and domain specific functioning score covering the following domains:

- Cognition – understanding & communicating
- Mobility– moving & getting around
- Self-care– hygiene, dressing, eating & staying alone
- Getting along– interacting with other people
- Life activities– domestic responsibilities, leisure, work & school
- Participation – joining in community activities, participating in society

WHO DAS scoring takes into account multiple levels of difficulty for each WHODAS 2.0 item (i.e. IRT based scoring). This type of scoring for WHODAS 2.0 allows for more fine-grained analyses that make use of the full information of the response categories for comparative analysis across populations or subpopulations. It takes the coding for each item response as “none”, “mild”, “moderate”, “severe” and “extreme” separately, and then uses a computation to determine the summary score by differentially weighting the items and the levels of severity.

The scoring has three steps:

- Step 1 – Summing of recoded item scores within each domain.
- Step 2 – Summing of all six domain scores.
- Step 3 – Converting the summary score into a metric ranging from 0 to 100 (where 0 = no disability; 100 = full disability).

The WHO DAS 2.0 scoring syntax is listed online via the link shared below.

Key features of WHO-DAS 2.0 include:

- ICF derived clinical scale with well-established psychometric properties of validity, reliability and sensitivity to change over time.
- Cardinal measure which is applicable across all health conditions and allows to generate an overall and domain specific functioning scores.

- Established population norms.

The combination of this features sets WHO DAS 2.0 apart from other clinical instruments.

WHO DAS 2.0 can be used for assessment of functioning in multiple use cases:

- Clinicians can quantify the effectiveness of their intervention within their clinical population and with reference to the general population.
- Disability evaluators can determine disability status in fair, transparent, impartial and comparable way.
- Reimbursement experts can measure quality (functioning outcome) of Case-mix/DRG groups.

WHO DAS 2.0 has been developed for use in adult population and currently does not assess body impairments or environmental factors. Users who wish to asses impairments in body functions and can make use of the WHO Model Disability Survey (MDS)[1] items which are also included in the function assessment part of the V Chapter.

WHO DAS 2.0 versions in three different formats (i.e. clinician administered, self-administered and proxy-administered) can be found via the link below.

WHO DAS 2.0 manual provides practical guidance for administering and scoring WHO DAS 2.0. including information on the simple and complex (Item-Response-Theory, IRT) scoring algorithm and population norms for the WHO DAS 12 and 36 item version. The WHO DAS 2.0 manual can be downloaded from the WHO website link - WHO Disability Assessment Schedule (WHODAS 2.0) - <https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health/who-disability-assessment-schedule> together with other languages version of the WHO DAS 2.0.

[1] The MDS is a ICF based general population survey developed by WHO and the World Bank that provides detailed and nuanced information on the lives of people with disability. It allows direct comparison between groups with differing levels and profiles of disability.

2.11.1.2 WHO DAS 2.0: representation and coding structure

To enable easy information retrieval each WHO DAS 2.0 item has an assigned alphanumeric code to represent the captured concepts of the respective item. The code starts with the letters “VD” followed by a randomly assign number. Additional item specific information and values (i.e. item question, five-point response scale) are included in the ICD-11 APIs to allow for easy integration of the instrument and the automated scoring computation in any software application (e.g. integration of WHO DAS 2.0 as health outcome measure in EMR software). The six WHO DAS 2.0 domains under which the 36 items of the instrument are grouped are represented as block heading. Like other ICD-11 block headings they are assigned an URI but not a code. Table 1 below shows an example of how a WHO DAS 2.0 item are represented.

WHO DAS 2.0 item	ICD-11 Code	URI
Attention functions	VD00	http://id.who.int/icd/entity/532448599
Item description question	Item response scale value set	Mapped ICF category
Because of your health condition, in the past 30 days, how much difficulty did you have in concentrating on doing something for ten minutes?	1=no difficulty, 2=mild difficulty, 3=moderate difficulty, 4=severe difficulty, 5=complete difficulty	d160 Focusing attention

2.11.2 Generic functioning entity

2.11.2.1 Functioning entities: features and use cases

The main tool for documenting functioning is a list of generic functioning domains derived from the ICF Annex 9 and the ICF generic core set.

The generic list allows clinicians to generate a coded functioning profile of an individual which can be used for multiple purposes like - goal setting - monitoring change - communicating across the continuum of care.

Because the list is directly linked to the ICF, it may - Facilitate joint use of ICD & ICF (code once – use multiple times) - Serve as an entry point the use of the full ICF.

2.11.2.2 Functioning entity: representation and coding structure

In accordance with the WHO FIC content model a functioning entity has two components i.e. category and a qualifier. Both have an assigned alphanumeric code and an URI to represent the captured concepts. The code starts with the letters “VV” followed by a randomly assign number. The functioning category and the qualifier are linked through postcoordination.

2.12 Electronic recording and reporting

Electronic documentation will follow the principle of lossless collection of information at the source. Best practice includes reporting of:

1. A text field that captures the clinical term or cause of death with the exact wording reported by the health provider, and
2. A data field that retains the identifier (URI) of the chosen entity of ICD-11 (index, code title or other element) that represents the most exact match for that text.
3. A data field for the relevant ICD-11 code.

In this way, the quality of the coding can be verified at any point in time. Also, specific conditions can be identified and analysed, independently of them being linked to an individual ICD code or lumped together in a code with other conditions.

Coding shall be done using tools based on the ICD-11 API, like the ICD-11 coding tool. Software must not include lists or other prompts to guide the recording or coding, as these

necessarily limit the range of diagnoses and therefore have negative impact on the accuracy and usefulness of the record and report.

2.13 Foundation Component and Tabular lists

The Foundation Component is a multi-dimensional collection of all ICD entities. Entities can be diseases, disorders, injuries, external causes, signs and symptoms. Some entities may be very broad, for example 'Injuries to the elbow or forearm', while others are more detailed, for example 'Fracture of upper end of ulna, extending into joint'. The Foundation Component also has the necessary information to use the entities to build a tabular list. The Foundation Component includes information on where and how a certain entity is represented in a tabular list, whether it becomes a grouping, a category with a stem code, or whether it is mentioned as an inclusion term in a particular category.

Several different tabular lists can be built from the Foundation Component. Drawing on the same Foundation Component, a set of tabular lists that builds on the same hierarchical tree structure can be created – producing congruent tabular lists. The Foundation Component includes instructions on how to combine certain codes in a tabular list to achieve more detail in coding. These rules help coders and computer systems to visualise the permitted code combinations when they are using a tabular list.

Core tabular lists for international use:

- Mortality and Morbidity Statistics (MMS)
- Primary care low resources settings (PCL)
- Verbal Autopsy (VA)
- Startup Mortality List (SMoL)

The full name of such a tabular list will always include 'ICD-11', e.g. ICD-11 MMS.

In a Tabular list, entities of the Foundation Component become categories. The categories are mutually exclusive and jointly exhaustive and linked to a mono hierarchical tree (they have only one parent). The information related to an entity that has become a category and has multiple parents is still available from the Foundation. This information can be used to visualise that category in more than one place in the Tabular list, e.g. showing them in black in its place for reference tabulation and in grey in any other place for browsing or alternative tabulations. ICD-11 has multiple congruent tabular lists with varying levels of detail.

The Foundation Component is also the data source for production and maintenance of Tabular lists, index and the Reference Guide. It also includes additional content (see 'content model') that goes beyond the traditional paper-based use of a classification. Depending on the setting within a country, it may be decided to use the full Foundation Component or to focus on the parts that are essential to production and maintenance of the Index and the Tabular list.

The Foundation Component serves to align the content of the different tabular lists and to define their categories. As such it allows standardised use of the ICD-11, independent of the setting in which it is used. The Foundation Component includes, for example, links to other classifications or terminologies that can be expanded in the future. Please note that the

mention of a term or entity in the Foundation exclusively serves ontological purposes. Mention of a term or entity in the Foundation does not mean approval or endorsement of a particular condition.

2.14 Main uses of the ICD: Mortality

This section concerns the rules and guidelines adopted by the World Health Assembly regarding the selection of a single cause or condition for routine tabulation from death certificates. Guidelines are also provided for the application of the rules and for coding of the condition selected for tabulation. Implementation of the ICD for mortality requires setting up an infrastructure for reporting and storing information, designing information flows, quality assurance and feedback, and training for classification users working with the input or output of data.

Following the introductory information in this Section and [2.15](#), Section [2.16](#) explains the basic concepts used in mortality coding. Sections [2.17](#) - [2.21](#), supplemented by Annexes in Section [3.14](#), guides how to code and identify the underlying cause of death, and Section [2.25](#) explains descriptions used in statistical tabulation and international reporting for mortality.

2.15 Mortality statistics

Mortality statistics are widely used for medical research, monitoring of public health, evaluating health interventions, allocation of health resources and planning and for, and follow-up of health care. Analysis of mortality data typically involves comparisons of data sets, for example those representing different geographical regions or different points in time. Unless the data have been produced by the same methods and according to the same standards, such comparisons will yield misleading results.

To standardise production of mortality data, WHO issues international instructions on data collection, coding and classification, and statistical presentation of causes of death. It is of utmost importance that production of mortality data follows the procedures detailed next, since any deviation from the international instructions will impair international comparability. The description of a single underlying cause of death, and selected approaches to capture further information on causes of death also reported on a certificate, enables the identification of trends in health for a given population. The following sections contain information on coding causes of death for mortality statistics. They explain the basic concepts, how to code conditions reported on death certificates, and how to select and tabulate the underlying cause of death.

The aim of these instructions is to optimise the mortality statistics from a public health point of view. Some of the instructions may appear wrong or questionable from a purely medical perspective. They should still not be set aside, since they may be motivated by well-founded epidemiological and public health principles. If an apparent error is found, it should be reported to WHO through the online proposal mechanism. WHO will either explain the rationale or take steps to correct the error at the international level. Individual countries should not correct what is assumed to be an error, since changes at the national level will lead to data that are less comparable to data from other countries, and thus less useful for analysis.

2.15.1 What is tabulated: Underlying cause of death

Effective public health interventions prevent harm or death by breaking the chain of events that lead to harm. For this purpose, the underlying cause of death has been defined as ‘(a) the disease or injury which initiated the train of morbid events leading directly to death, or (b) the circumstances of the accident or violence which produced the fatal injury’, and is selected for routine single-cause tabulation of mortality statistics. See Sections [2.17.2 Selecting the underlying cause of death](#) and [2.17 Coding instructions for mortality](#) for specific coding instructions to identify the underlying cause of death.

2.15.2 Data source: The international form of Medical Certificate of Cause of Death (MCCD)

The international mortality coding instructions presuppose that data have been collected with a death certificate conforming to the international form of Medical Certificate of Cause of Death as recommended by the WHO. Frame A, the medical data part of the international form is split into two parts: Part 1 is for diseases related to the chain of events directly leading to death, and Part 2 is for other significant conditions contributing to death. Other information in the form is also used in identifying the underlying cause of death for tabulation.

In order to align the way this information is collected internationally, the form should be followed as closely as possible. Otherwise, the causes of death cannot be coded and selected according to the international standard and the data will not be internationally comparable. For example, some coding instructions apply to conditions reported as caused by certain other conditions, and in such cases, it is important to have a clear distinction between causes reported in Part 1 and in Part 2 of the death certificate. Further, information reported elsewhere on the certificate, such as manner of death or whether pregnancy contributed to the death, is essential when assigning multiple cause codes to the conditions stated on the certificate and selecting an underlying cause for tabulation.

It is the responsibility of the medical practitioner or other qualified certifier completing the medical certificate of cause of death (MCCD) to use their clinical judgment to indicate which morbid conditions led directly to death and to state any pre-existing conditions giving rise to this cause. Electronic certification systems should not include predetermined lists of causes of death to guide the certifier, as these necessarily limit the range of diagnoses and therefore have an adverse effect on the accuracy and usefulness of the report.

Administrative Data (can be further specified by country)																			
Sex	☐ Female				☐ Male				☐ Unknown										
Date of birth	D	D	M	M	Y	Y	Y	Y	Date of death	D	D	M	M	Y	Y	Y	Y		
Frame A: Medical data: Part 1 and 2																			
1 Report disease or condition directly leading to death on line a Report chain of events in due to order (if applicable) State the underlying cause on the lowest used line	  	a	Cause of death	Time interval from onset to death															
		b	Due to:																
		c	Due to:																
		d	Due to:																
2 Other significant conditions contributing to death (time intervals can be included in brackets after the condition)																			
Frame B: Other medical data																			
Was surgery performed within the last 4 weeks?										☐ Yes		☐ No		☐ Unknown					
If yes please specify date of surgery										D	D	M	M	Y	Y	Y	Y		
If yes please specify reason for surgery (disease or condition)																			
Was an autopsy requested?										☐ Yes		☐ No		☐ Unknown					
If yes were the findings used in the certification?										☐ Yes		☐ No		☐ Unknown					
Manner of death:																			
☐ Disease					☐ Assault					☐ Could not be determined									
☐ Accident					☐ Legal intervention					☐ Pending investigation									
☐ Intentional self harm					☐ War					☐ Unknown									
If external cause or poisoning:										Date of injury		D	D	M	M	Y	Y	Y	Y
Please describe how external cause occurred (If poisoning please specify poisoning agent)																			
Place of occurrence of the external cause:																			
☐ At home			☐ Residential institution			☐ School, other institution, public administrative area					☐ Sports and athletics area								
☐ Street and highway			☐ Trade and service area			☐ Industrial and construction area					☐ Farm								
☐ Other place (please specify):										☐ Unknown									
Fetal or infant Death																			
Multiple pregnancy										☐ Yes		☐ No		☐ Unknown					
Stillborn?										☐ Yes		☐ No		☐ Unknown					
If death within 24h specify number of hours survived												Birth weight (in grams)							
Number of completed weeks of pregnancy												Age of mother (years)							
If death was perinatal, please state conditions of mother that affected the fetus and newborn																			
For women, was the deceased pregnant?										☐ Yes		☐ No		☐ Unknown					
☐ At time of death										☐ Within 42 days before the death									
☐ Between 43 days up to 1 year before death										☐ Unknown									
Did the pregnancy contribute to the death?										☐ Yes		☐ No		☐ Unknown					

The 2016 International form of the Medical Certificate of Cause of Death

See [3.14 Annex C: Annexes for Mortality Coding](#) for further details and guidance.

2.15.3 Routine use and special cases

2.15.3.1 Routine cause of death reporting systems

In routine cause of death reporting systems, every individual death is certified by a qualified medical doctor who carries out an accurate postmortem examination, collects history from relatives, and has access to all pre-existing medical information about the defunct. The medical certification of the cause of death is usually the responsibility of the attending physician and should be in line with international recommendations. Administrative procedures should ensure confidentiality of data from death certificates or other medical records.

In the case of deaths certified by coroners or other legal authorities, the medical evidence supplied to the certifier should be stated on the certificate in addition to any legal findings.

Routine cause of death reporting is usually embedded in the certification of death process. Death certificates are a legal requirement for burial and for inheritance.

2.15.3.2 Verbal autopsy

Verbal autopsy (VA) is a method used to ascertain the cause of a death based on an interview with next of kin or other caregivers where no medical certification is available. This is done using a standardised instrument that elicits information on signs, symptoms, medical history, and circumstances preceding death. The cause of death, or the sequence of causes that led to death, are assigned based on the data collected by the instrument and other available information. Rules and guidelines, algorithms or computer programs, may assist in evaluating the information to determine the cause of death.

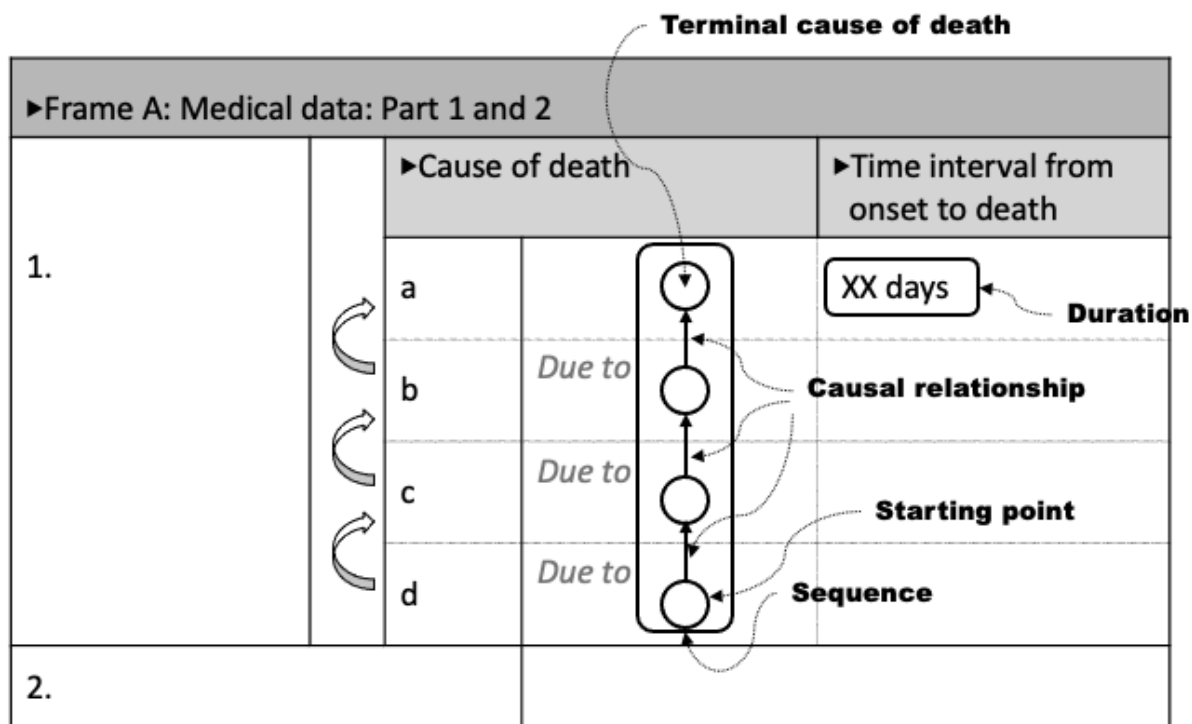
The main objective of the VA is to describe the causes of death at the community or population level in areas, where civil registration and death certification systems are weak and where most people die at home without having had contact with the health system. A standard VA instrument comprises a VA questionnaire, cause of death or mortality classification system, and diagnostic criteria (either expert or data derived algorithms) for deriving causes of death.

The VA process consists of interviews, data recording, and identification of the cause of death from the reports. At any step, factors can influence the cause-specific mortality fractions estimated throughout the process. Besides research, VA is a viable method for causes of death identification in settings where no physician can evaluate the deceased. More information can be found on the [WHO Verbal autopsy webpage](#).

See [3.14.14 Target list of causes of death for verbal autopsy](#) with corresponding ICD-11 codes.

2.16 Basic concepts

Mortality coders must be familiar with the basic concepts introduced in this section.



Basic Concepts

Basic concepts for mortality coding in the international form of the Medical Certificate of Cause of Death

2.16.1 Terminal cause of death

The disease or condition entered first on the first used line of Part 1 of the death certificate is the cause directly leading to death. This is known as the terminal or immediate cause of death.

Example 1

- 1 (a) Myocardial infarction
due to
- (b) Coronary atherosclerosis
due to
- (c)
due to
- (d)

2

The myocardial infarction is the terminal cause of death, since it is entered first on the first used line of the certificate.

Example 2

- 1 (a) Myocardial infarction and pulmonary oedema
due to
 - (b) Coronary atherosclerosis
due to
 - (c)
due to
 - (d)
- 2

The myocardial infarction is the terminal cause of death, since it is entered first on the first used line of the certificate.

2.16.2 Causal relationship and sequence

A causal relationship exists if a condition mentioned on the death certificate can be caused by another condition also mentioned on the certificate. The term 'sequence' refers to a chain or series of medical events in which each step is a complication of, or is caused by, the previous step. A causal relationship can exist between any two conditions, including conditions assigned to the same ICD-11 code, regardless of where each condition was reported. In a correctly completed death certificate, a sequence is a set of conditions reported line by line with a causal relationship between each element. Four lines are provided in Part 1 of the certificate for recording the sequence of events leading to the death. A certifier may only use as many lines as are required to describe the sequence.

A causal relationship is considered acceptable for mortality coding if it is founded not only on a medical assessment but also on epidemiological and public health considerations. Therefore, a medically acceptable relationship might be listed as unacceptable in the coding instructions because a later step in the sequence is more important from a public health point of view.

In addition, a reported sequence that appears improbable should be accepted if one or more intervening steps would explain the causal relationship, even if these have not been reported. However, such assumed intervening causes are not to be coded, as they are assumptions and not reported conditions.

To decide whether a stated causal relationship is acceptable, always apply the instructions in Section [2.19.1 Special instructions on accepted and rejected sequences \(Steps SP3 and SP4\)](#). Stated relationships that are not listed in Section [2.19.1](#) should be accepted as far as possible, because the certifier's opinion about the causes leading to death should not be disregarded.

Example 1

- 1 (a) Myocardial infarction
due to
 - (b) Coronary thrombosis
due to
 - (c) Coronary atherosclerosis
due to
 - (d)
- 2

The terminal cause of death is myocardial infarction. It is caused by the coronary thrombosis, which, in turn, is caused by coronary atherosclerosis. Consequently, the sequence is: myocardial infarction due to coronary thrombosis due to coronary atherosclerosis.

Example 2

- 1 (a) Extensive haemorrhage
due to
 - (b) Traumatic amputation of right leg
due to
 - (c) Run over by a bus
due to
 - (d)
- 2

The terminal cause of death is haemorrhage. It is a complication of the traumatic amputation of the right leg, which, in turn, is caused by the bus accident. Consequently, the sequence is: extensive haemorrhage due to traumatic amputation of the right leg due to being run over by a bus.

2.16.3 Starting point

The starting point is the condition or event that started the sequence of acceptable causal relationships ending with the terminal cause of death. In a correctly completed death certificate, the condition reported first on the lowest used line in Part 1 is the starting point of the sequence. The instructions on how to identify the starting point is provided in [Section 2.17.3 Find the starting point \(Steps SP1 to SP8\)](#).

If the death certificate is not correctly filled out, the starting point may be reported elsewhere, and instructions are given to identify the starting point also for such cases in a standardised manner. Therefore, it is important to apply the instructions in [Section 2.17.3](#) in a sequential manner.

The condition provisionally considered as the starting point when applying the instructions step by step is referred to as the 'tentative starting point (TSP)' and may change several times as the instructions are applied to the death certificate.

Example 1

- 1 (a) Myocardial infarction and pulmonary oedema
due to
 - (b) Coronary atherosclerosis
due to
 - (c)
due to
 - (d)
- 2

Coronary atherosclerosis is the starting point, since it led to the myocardial infarction.

Example 2

- 1 (a) Pneumonia
due to
 - (b) Hip fracture
due to
 - (c) Tripped on carpet
due to
 - (d)
- 2

Tripped on carpet is the starting point, since it started the sequence of events leading to death.

2.16.4 Duration

On death certificates, each reported condition should also include information about duration. The duration refers to the interval from the onset of the disease or condition to the time of death. Note that it is not always the same as the time of diagnosis of the condition, which may be at the same time as, or after, the onset of symptoms.

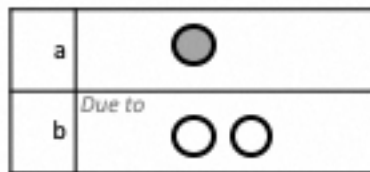
2.16.5 First-mentioned sequence

A death certificate may contain several sequences of acceptable causal relationships ending with the terminal cause of death. The coding instructions are given to identify the starting point of the first-mentioned sequence in Part 1 (See also Step SP4).

The figures below illustrate examples of certificates where each condition reported is shown by a circle. The starting point of the first-mentioned sequence is in grey, and the causal relationship of the first-mentioned sequence is indicated by an arrow.

To identify the first-mentioned sequence, begin with the terminal cause of death (the condition entered first on the first used line of Part 1). Check if the conditions on the next line in Part 1 can result in the terminal cause of death. If several conditions are reported on the same line, check from left to right in turn until you find a condition that could cause the terminal cause.

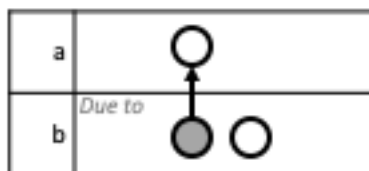
If no condition on the next line can cause the terminal cause of death, there is no sequence ending with the terminal cause of death. The terminal cause of death is the tentative starting point. Specific instruction is given also when you find no sequence (see Step SP5).



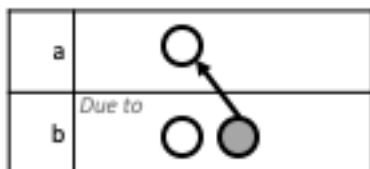
NoSequence

No Sequence ending with the terminal cause of death

If there is a condition that can cause the terminal cause of death, the first condition found to be able to cause the terminal cause of death is the tentative starting point. If there are no conditions reported on lower lines, the sequence between this tentative starting point and the terminal cause is the first-mentioned sequence.



First Mentioned Sequence A



First Mentioned Sequence B

If there are conditions reported on lower lines in Part 1, repeat the procedure for the next lower line. Start with the tentative starting point identified in the previous step. Check the conditions on the next lower line in Part 1, from left to right, to determine if they can cause the tentative starting point. Continue until you have found a condition that can cause the tentative starting point. This is the new tentative starting point.